GRIEVANCE360

A Smart Grievance Redressal and Analytics Portal
for Educational Institutions

MODULE DESCRIPTION

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Module Description

The system comprises thirteen functional modules. Modules M1 to M7 implement the core grievance lifecycle. Modules M8 to M13 extend the system with administration, escalation, duplicate handling, self service, announcements and performance measurement. Each module is described below with its purpose, the functions it provides, its inputs and outputs, and the business rules it enforces.

Table 1 Module summary

Code	Module	Primary user	Routes	Tables used
M1	Authentication and Access Control	All	7	users
M2	Complaint Submission	Student	1	complaints, categories, history
M3	Complaint Tracking and Interaction	Student	5	complaints, history, comments, feedback
M4	Assignment and Auto Routing	Administrator	1	complaints, departments, users
M5	Complaint Processing	Staff	2	complaints, history
M6	Notification and Live Update	All	3	notifications
M7	Analytics and Reporting	Administrator	5	complaints, categories, departments, feedback
M8	Administration Console and Master Data	Administrator	9	users, departments, categories
M9	Escalation and Reminder Engine	Administrator	3	escalations, complaints, notifications
M10	Duplicate Detection and Grouping	Student, Administrator	4	complaint_links, complaint_supporters
M11	Knowledge Base and Self Service	Student	5	kb_articles
M12	Announcements and Notice Board	Administrator	4	announcements
M13	Staff Performance Scorecard	Administrator	2	complaints, feedback, users

Table 2 Scale of the module set

Measure	Core (M1 to M7)	Extension (M8 to M13)	Total
Modules	7	6	13
Application routes	24	27	51
Database tables	8	5 additional	13
Documented algorithms	3	3	6

1. Module M1 - Authentication and Access Control

Purpose

Establishes identity, maintains the session, and guarantees that each user reaches only the areas and records to which they are entitled.

Functions provided

Table 3 Functions of module M1

Function	Route	Description
Landing page	/	Public entry point. Shows headline counts and redirects an already authenticated user to their own dashboard.
Registration	/register	Self service student sign up. Validates name length, mail address format, minimum password length and password confirmation, and rejects an address already present in the users table.
Login	/login	Verifies the submitted password against the stored hash and establishes a session.
Role redirect	/home	A single decision point that sends each role to its own landing page, so no other route needs to repeat that logic.
Logout	/logout	Terminates the session.
Forgot password	/forgot	Issues a 32 byte URL safe token with a one hour expiry and stores it against the user record.
Reset password	/reset/token	Validates the token and its expiry, then replaces the password hash and clears the token so that it cannot be reused.

Business rules enforced

- Passwords are never stored in plain text. They are hashed with Werkzeug on registration and on reset, and only ever compared as hashes.
- A failed login returns a single generic message regardless of whether the mail address exists, so the form cannot be used to enumerate registered accounts.
- The forgot password page returns the same acknowledgement for a registered and an unregistered address, for the same reason.
- A reset token is single use and expires after one hour.
- A deactivated account cannot log in even with correct credentials.
- The role_required decorator is applied to every route that is not public.

Authorisation model

Table 4 Permissions by role

Role	May view	May modify
Student	Only complaints they raised	Own complaints: add comment, submit rating, reopen
Staff	Complaints assigned to them or routed to their department	Assigned complaints: update status with remark, add comment
Administrator	All complaints and all analytics	Assign or reassign any complaint, update any status

Role checks alone are insufficient. A student holds the correct role for the tracking page, but must not open another student's ticket. The application therefore performs a second, record level ownership check through can_view on every route that addresses a specific complaint.

2. Module M2 - Complaint Submission

Purpose

Captures a new grievance and establishes every derived value on which the rest of the system depends, namely the ticket code, the resolution deadline and the initial priority score.

Input fields

Table 5 Submission form fields and validation

Field	Type	Validation
Category	Dropdown, required	Must match an existing category row
Subject	Text, required	Minimum 5 characters, maximum 200
Description	Textarea, required	Minimum 15 characters
Attachment	File, optional	Extension must be png, jpg, jpeg, gif, pdf or docx, maximum 5 MB
Anonymous	Checkbox, optional	When set, staff and administrator views display Anonymous in place of the student name

Processing sequence

1. Validate all fields and collect every error before responding, so that the user is not corrected one field at a time.
2. Validate and store the attachment if one is present. The stored filename is prefixed with twelve random hexadecimal characters, so two users uploading a file of the same name cannot overwrite one another.
3. Read sla_days from the selected category and compute the deadline as the submission time plus that many days.
4. Compute the initial priority score from the category weight.
5. Insert the complaint row, then derive the ticket code from the assigned row identifier and write it back, which guarantees that the code and the record always correspond.
6. Write the first audit row with a null previous status and the new status Pending.
7. Insert a notification for every administrator.

Ticket code format

```
GRV - <four digit year> - <four digit sequence>

Example:  GRV-2026-0031
```

3. Module M3 - Complaint Tracking and Interaction

Purpose

Gives the student continuous visibility of their complaint and the means to interact with it after submission.

Functions provided

Table 6 Functions of module M3

Function	Route	Description
Dashboard	/student	All complaints raised by the logged in student, rendered as responsive cards with status badges and SLA state, together with four summary tiles.
Track	/complaint/code	Full detail, attachment link, vertical status timeline built from the audit table, comment thread, and the rating or reopen controls where applicable.
Comment	/complaint/id/comment	Appends a message to the thread and notifies the other party, never the author.
Feedback	/complaint/id/feedback	Records a rating of 1 to 5 with an optional remark.
Reopen	/complaint/id/reopen	Returns a completed complaint to active handling.

Conditional interface rules

- The rating form is rendered only when the status is Resolved or Closed, and only for the student who raised the complaint. The server rechecks both conditions on submission, so hiding the form is a convenience rather than the security control.
- A rating may be submitted only once per complaint, which is enforced by a unique constraint on complaint_id in the feedback table.
- Submitting a rating transitions the complaint to Closed and records that transition in the audit table.
- The reopen control appears only for a complaint in the Resolved, Closed or Rejected state.
- Reopening increments reopen_count, clears resolved_at, recalculates the deadline from the reopen moment, and raises the priority score.

4. Module M4 - Assignment and Auto Routing

Purpose

Determines which department is responsible for a complaint and which member of staff will act on it.

Auto routing mechanism

Each department carries a keyword vocabulary. When the administrator opens the assignment screen, the complaint subject and description are scanned against every vocabulary and each department receives a score. A keyword found in the subject contributes three points; the same keyword found in the description contributes one point. The department with the highest total is presented as the suggestion.

```
for each department:
    score = 0
    for each keyword in that department's vocabulary:
        if whole-word match in subject:      score += 3
        if whole-word match in description:  score += 1
suggestion = department with the highest score
if every score is zero: no suggestion is offered
```

Matching uses whole word boundaries rather than substring matching. Without that constraint the keyword ac, standing for air conditioning, would match inside the word academics and route maintenance complaints to the academic office.

When no keyword matches at all, the system deliberately offers no suggestion and asks the administrator to choose. Presenting a confident but arbitrary guess is worse than presenting none, because an administrator working quickly will accept it.

Assignment support

- The staff dropdown displays each member's current open complaint count, so that work can be distributed rather than concentrated.
- Selecting a department filters the staff list to that department's members.
- Assignment sets the department, the assignee and the status Assigned, writes an audit row naming the assignee, and notifies both the assigned member of staff and the student.

This is a rule based classifier and not machine learning. That is a considered choice. Its decisions can be explained line by line, it requires no training data, it cannot degrade unpredictably, and the keyword lists can be tuned by non programmers as institutional vocabulary changes. For a fixed set of departments and a stable problem domain it is the appropriate tool.

5. Module M5 - Complaint Processing

Purpose

Provides staff with their work queue and the means to advance a complaint through the workflow.

Status vocabulary

Table 7 Complaint statuses

Status	Meaning	Set by
Pending	Submitted, not yet assigned	System on submission
Assigned	Allocated to a member of staff	Administrator
In Progress	Actively being worked	Staff
Resolved	Work complete, awaiting confirmation	Staff
Closed	Confirmed complete	Student rating, or staff
Reopened	Returned to handling after resolution	Student
Rejected	Outside scope or not reproducible	Staff

Processing rules

- The staff queue is sorted by live priority score in descending order, so the most urgent complaint is always presented first.
- A remark is mandatory on every status change. The remark is displayed to the student on the timeline and forms the permanent record of what was done.
- Every change writes an audit row capturing the previous status, the new status, the remark, the responsible user and the timestamp.
- Transitions to Resolved or Closed set resolved_at, which is the measurement point for the average resolution time analytic.
- The student is notified on every change.

6. Module M6 - Notification and Live Update

Purpose

Delivers state changes to the user without requiring a page reload.

Endpoints and clients

Table 8 Live update endpoints

Endpoint	Client	Interval	Behaviour
/api/complaint/code/status	track.js	5 seconds	Returns the current status, colour, deadline, breach flag, days remaining and the audit row count.
/api/notifications	notify.js	15 seconds	Returns the ten most recent notifications and the unread count.
/api/notifications/read	notify.js	On demand	Marks all of the user's notifications as read.

Update strategy

The tracking client updates the status badge, the deadline and the SLA line in place. It does not attempt to rebuild the timeline in JavaScript. Instead it compares the audit row count returned by the interface against the count present when the page was rendered, and when the server count is higher the page is reloaded once so that the server rebuilds the timeline.

This avoids maintaining two independent renderings of the same data, one in Jinja and one in JavaScript, which would inevitably diverge. The frequent and inexpensive update is handled on the client, while the rare and complex update is delegated back to the server.

Security of client rendering

Notification text incorporates complaint subjects, which are user input. Before insertion into the document the client assigns the text to a textContent property and reads back the escaped innerHTML, so that markup contained in a subject is rendered as literal characters and cannot execute.

7. Module M7 - Analytics and Reporting

Purpose

Converts the complaint record into management information, and exports it.

Analytics provided

Table 9 Analytics and their derivation

Analytic	Method	Presentation
Distribution by status	GROUP BY on status	Doughnut chart, with colours matched to the status badges used throughout the interface
Distribution by category	LEFT JOIN with GROUP BY	Horizontal bar chart, so that long category names remain readable on a narrow screen
Distribution by department	LEFT JOIN with GROUP BY	Vertical bar chart
Monthly trend	Aggregated in Python by year and month	Line chart
Average resolution time	Difference between the submission and resolution timestamps, averaged per department in Python	Sorted table showing mean days and the number of resolved complaints contributing
SLA breach report	Filtered query on status and deadline	Table sorted by deadline, oldest breach first
Top recurring issue	Highest category count	Highlighted banner identifying the largest single source of grievances
Satisfaction	AVG and COUNT over feedback	Summary tiles

Average resolution time and the monthly trend are both computed in Python rather than in SQL. Both require date arithmetic. MySQL provides TIMESTAMPDIFF and DATE_FORMAT while SQLite provides julianday and strftime, so using either would break portability and force two versions of the query. Performing the arithmetic in Python keeps a single implementation that runs correctly on both engines, and the aggregation covers resolved complaints only, so the volume is modest.

Search, filtering and export

- Filters cover free text search across ticket code, subject, description and student name, together with status, category, department, a date range and a breached only switch.
- Results are paginated at twenty per page.
- The Excel export writes a genuine .xlsx workbook with a styled header row, sized columns and a frozen header pane.
- The PDF export renders a print optimised A4 landscape report page and uses the browser print dialogue, which keeps the application free of additional rendering dependencies.
- Both exports reuse the same filter parsing function as the list screen, so the exported rows are exactly the rows on screen. There is no second implementation of the filter logic that could drift.

8. Module M8 - Administration Console and Master Data

Purpose

Allows the administrator to manage the reference data and the user accounts on which every other module depends, without editing source code or reinitialising the database. Before this module existed, adding a member of staff or a new complaint category required a developer and a database rebuild, which destroyed all existing records.

Functions provided

Table 10 Functions of module M8

Function	Route	Description
User list	/admin/users	All accounts with role and department, searchable by name or mail address and filterable by role and active state.
Add user	/admin/users/new	Adds a staff or administrator account, assigns a department and sets an initial password, which is stored only as a hash.
Edit user	/admin/users/id/edit	Changes name, mail address, telephone number, role and department.
Activate or deactivate	/admin/users/id/toggle	Suspends an account without deleting it, so the historical record of what that user did remains intact.
Reset password	/admin/users/id/password	Sets a new password for a user who cannot access their own reset link.
Department list	/admin/departments	Lists departments with the count of complaints routed to each.
Edit department	/admin/departments/id/edit	Renames a department or changes its contact address.
Category list	/admin/categories	Lists categories with their service level in days and priority weight.
Edit category	/admin/categories/id/edit	Changes the service level, the priority weight or the owning department.

Business rules enforced

- An administrator cannot deactivate their own account or reduce their own role, which prevents the last administrator from locking everyone out of the system.
- A department or category that is referenced by an existing complaint cannot be deleted. It may be renamed, so historical records keep a meaningful label.
- Changing the service level of a category applies only to complaints submitted afterwards. Existing deadlines are never rewritten, because a deadline that moves is not a deadline.

- Deactivating a member of staff who holds open complaints raises a warning listing those complaints, so that they can be reassigned first.
- A password set by an administrator is stored only as a hash, exactly as one chosen by the user.
- Every action in this module writes to the audit record, because master data changes affect the behaviour of the whole system.

9. Module M9 - Escalation and Reminder Engine

Purpose

Ensures that a complaint which is not acted upon rises through levels of authority automatically. In every manual process the burden of chasing an unanswered request falls on the student, which is precisely the person least able to apply pressure. This module transfers that burden to the system.

Escalation levels

Escalation is driven by the proportion of the service level that has elapsed, rather than by a fixed number of days, so a category with a two day service level escalates faster than one with ten days.

```
elapsed_ratio = (now - submitted_at) / (deadline - submitted_at)
```

Table 11 Escalation levels and their triggers

Level	Trigger	Action taken
0 Normal	Below 0.5	No action
1 Reminder	0.5 and above	Reminder notification to the assigned member of staff
2 Department	1.0 and above, that is past the deadline	Notification to every administrator, complaint flagged as breached
3 Management	2.0 and above, twice the allowed time	Flagged for management review and raised to maximum priority band

Functions provided

Table 12 Functions of module M9

Function	Route	Description
Escalation dashboard	/admin/escalations	Lists complaints by escalation level, oldest first, with the level history of each.
Run sweep	/admin/escalations/run	Evaluates every open complaint, raises the level where a threshold has been crossed and issues the corresponding notifications.
Escalation check	/api/escalation/check	Returns the count at each level, used to keep the administrator dashboard current.

Business rules enforced

- A complaint may only move up a level, never down, while it remains open. This prevents a complaint oscillating across a threshold and issuing repeated notifications.
- One record is written per complaint per level, so running the sweep twice produces no duplicate notifications. The operation is safe to repeat.
- Reopening a complaint resets its escalation level to zero, because the deadline is recalculated from the reopen moment.
- A complaint in the Resolved, Closed or Rejected state is never escalated.
- Every level change is recorded with the reason and the users notified, so the escalation history can be shown alongside the status timeline.

Table introduced

Table 13 Structure of the escalations table

Column	Type	Description
id	INT	Primary key
complaint_id	INT	Foreign key to complaints, indexed
from_level	INT	Level before the change
to_level	INT	Level after the change
reason	TEXT	Threshold crossed and the elapsed proportion
notified_user_id	INT	Foreign key to users, nullable
escalated_at	DATETIME	When the level changed

10. Module M10 - Duplicate Detection and Grouping

Purpose

Identifies complaints that describe the same underlying problem and allows them to be handled as one. When a network switch fails, forty students may each submit a separate complaint. Without this module the institution receives forty tickets and forty separate resolution efforts, and no indication that they are a single fault.

Similarity measure

Similarity is measured between the word sets of two complaints using the Jaccard coefficient, which is the size of the intersection divided by the size of the union. Comparison is restricted to complaints in the same category that are still open, which keeps the comparison inexpensive.

```
A = distinct significant words of complaint one
B = distinct significant words of complaint two

similarity(A, B) = | A intersect B |  /  | A union B |

words shorter than four characters and common words are removed
a match on the subject line counts twice
candidate is offered when similarity >= 0.35
```

The Jaccard coefficient was chosen over a simple count of shared words because a long complaint would otherwise appear similar to everything. Dividing by the union normalises for length, so a short complaint and a long one are compared fairly.

Functions provided

Table 14 Functions of module M10

Function	Route	Description
Similarity check	/api/similar	Called as the student types the subject. Returns open complaints in the same category above the similarity threshold.
Endorse an existing complaint	/complaint/id/metoo	Records the student as a supporter of an existing complaint instead of submitting a new one.
Merge	/complaint/id/merge	Administrator links a duplicate complaint to a parent. Every supporter of the child becomes a supporter of the parent.
Grouped view	/complaint/id/group	Shows the parent complaint together with every complaint linked to it and the full supporter list.

Business rules enforced

- A student may endorse a complaint only once, which is enforced by a unique constraint on the pair of complaint and user.
- A student may not endorse their own complaint.
- Supporter count raises the priority score, so a widely reported problem rises in the queue. The revised score adds three points per supporter.
- Merging is performed by an administrator, never automatically. The similarity measure advises; a person decides.
- Resolving a parent complaint resolves every complaint merged into it, and notifies every supporter at once.
- A merged complaint remains visible to the student who raised it, showing the parent it was joined to, so nobody's complaint silently disappears.
- Merging cannot form a chain. A complaint that is already a child cannot become a parent.

Revised priority score

The priority formula defined in module M5 is extended with a supporter term. A complaint reported by many students is genuinely more urgent than one reported by a single student, and the score now reflects that.

Tables introduced

Table 15 Structure of the complaint_links table

Column	Type	Description
id	INT	Primary key
parent_id	INT	The complaint that will be worked on
child_id	INT	The duplicate, unique so it cannot be merged twice
linked_by	INT	Administrator who performed the merge
linked_at	DATETIME	When the merge took place

Table 16 Structure of the complaint_supporters table

Column	Type	Description
id	INT	Primary key
complaint_id	INT	Foreign key to complaints, indexed
user_id	INT	The endorsing student
supported_at	DATETIME	When the endorsement was recorded

11. Module M11 - Knowledge Base and Self Service

Purpose

Answers a question before it becomes a complaint. A substantial proportion of grievances are requests for information that has already been given to somebody else, for example the procedure for applying for revaluation. Answering these from a searchable record costs nothing and returns an answer immediately rather than in several days.

Functions provided

Table 17 Functions of module M11

Function	Route	Description
Article list	/kb	Searchable list of published articles, filterable by category.
Article	/kb/id	Full article with a view counter and a helpfulness prompt.
Mark helpfulness	/kb/id/helpful	Records whether the article answered the question, which identifies articles that need rewriting.
Manage articles	/admin/kb	Administrator list with publish and unpublish controls.
Add or edit article	/admin/kb/save	Writes an article, optionally promoted directly from the remark on a resolved complaint.

Deflection at the point of submission

The submission form searches the knowledge base as the student types the subject, in the same request that checks for duplicate complaints. Matching articles are offered above the form. If the student opens an article and then abandons the submission, a deflection is recorded against that article.

The deflection count is the measure of whether this module is worth keeping. An article with many views and no deflections is not answering the question it claims to answer.

Business rules enforced

- Only a published article is visible to students. Draft articles are visible to the administrator alone.
- An article promoted from a resolved complaint carries no student identifying information, because the resolution is general even when the complaint was personal.
- A student may register helpfulness once per article.
- Articles are ordered by helpfulness and then by views, so the answers that actually work are found first.

Table introduced

Table 18 Structure of the kb_articles table

Column	Type	Description
id	INT	Primary key
title	VARCHAR(200)	Question as a student would phrase it
body	TEXT	The answer
category_id	INT	Foreign key to categories, nullable
is_published	BOOLEAN	Draft articles are hidden from students
views	INT	Number of times opened
helpful_yes	INT	Positive helpfulness responses
helpful_no	INT	Negative helpfulness responses
deflections	INT	Submissions abandoned after this article was opened
author_id	INT	Foreign key to users
published_at	DATETIME	When it became visible

12. Module M12 - Announcements and Notice Board

Purpose

Informs students of a known problem before they report it. When a fault is already known and being worked on, every further complaint about it consumes staff attention without adding information. An announcement shown at the point of submission removes that duplication at its source.

Functions provided

Table 19 Functions of module M12

Function	Route	Description
Notice board	/announcements	Active announcements for the logged in user, filtered by their role and department.
Manage announcements	/admin/announcements	Administrator list showing scheduled, active and expired entries.
Add or edit	/admin/announcements/save	Writes an announcement with an audience, a category and an active period.
Activate or deactivate	/admin/announcements/id/toggle	Withdraws an announcement immediately without deleting the record.

Targeting and display

- An announcement may be addressed to all users, or to one role, or to one department, or to a combination.
- An announcement carries a start and an end time and appears only within that period, so a planned maintenance notice can be written in advance.
- Active announcements appear as a banner on the dashboard of every user in the audience.
- An announcement may be linked to a complaint category. When a student selects that category on the submission form, the announcement is shown immediately with the option to abandon the submission.

Business rules enforced

- An announcement whose end time has passed is retained rather than deleted, so the record of what students were told remains available.
- The end time must be later than the start time.
- Deactivation is immediate and does not wait for the end time.
- Announcements are never shown to a user outside the stated audience, even by direct link.

Table introduced

Table 20 Structure of the announcements table

Column	Type	Description
id	INT	Primary key
title	VARCHAR(200)	Headline shown in the banner
body	TEXT	Full text
audience_role	VARCHAR(20)	all, student, staff or admin
department_id	INT	Restricts to one department, nullable
category_id	INT	Shown on the submission form for this category, nullable
starts_at	DATETIME	Beginning of the display period
ends_at	DATETIME	End of the display period
is_active	BOOLEAN	Allows immediate withdrawal
author_id	INT	Foreign key to users

13. Module M13 - Staff Performance Scorecard

Purpose

Measures how each member of staff and each department performs against the service level. Module M7 reports on complaints; this module reports on the people and offices handling them, which is the information required for a review meeting.

Metrics computed

Table 21 Performance metrics and their derivation

Metric	Derivation	Interpretation
Complaints handled	Count of complaints ever assigned to the member of staff	Volume, which gives context to every other figure
Average resolution time	Mean interval between assignment and resolution	Speed. Measured from assignment rather than submission, because a member of staff is not accountable for time before the complaint reached them
Service level compliance	Proportion resolved on or before the deadline, as a percentage	Reliability. The single most useful figure in the table
Average satisfaction	Mean student rating across their resolved complaints	Quality as judged by the person who raised the complaint
Reopen rate	Reopened complaints divided by resolved complaints, as a percentage	Whether work marked complete stayed complete
Currently open	Count of complaints not yet completed	Present workload
Currently overdue	Open complaints past their deadline	Immediate attention required

Functions provided

Table 22 Functions of module M13

Function	Route	Description
Scorecard	/admin/performance	One row per member of staff with every metric, sortable, together with a departmental summary.
Individual detail	/admin/performance/id	The same metrics for one member of staff, with the underlying complaint list and the rating remarks received.

Business rules and cautions

- All metrics are computed on read from the existing records. No aggregate is stored, so a figure can never disagree with the underlying data.
- Resolution time is measured from assignment rather than from submission, because a member of staff cannot be held responsible for the interval before the complaint was routed to them.
- A member of staff with fewer than five resolved complaints is marked as having insufficient data, because an average over two complaints is not a measurement.
- Satisfaction is reported alongside volume and never on its own, since a high average across three complaints means less than a moderate average across sixty.
- The scorecard is visible to the administrator only.